Final_Project_Tangavelou_VincentRMD

2026-10-07

#Data Preparation

```
meta_data <- read.csv("../Final Project/meta_data.csv")
eso_sub <- read.csv("../Final Project/eso_sub.csv")

library(janitor) #Used to clean data
```

```
## Warning: package 'janitor' was built under R version 4.4.3
```

```
##
## Attaching package: 'janitor'
```

```
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(tidyverse)   # dplyr, ggplot2, readr, tidyr, etc.
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tibble' was built under R version 4.4.3
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```
## Warning: package 'readr' was built under R version 4.4.3
```

```
## Warning: package 'purrr' was built under R version 4.4.3
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
## Warning: package 'stringr' was built under R version 4.4.3
```

```
## Warning: package 'forcats' was built under R version 4.4.3
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(haven)       # read_xpt()
```

```
## Warning: package 'haven' was built under R version 4.4.3
```

```
library(gridExtra)   # organizing plots
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(ggpubr)      # plot labels
library(ggplot2)    #plotting data
library(knitr) #Used to summarize table

t_eso_sub <-t(eso_sub)
t_eso_sub <- row_to_names(t_eso_sub, row_number = 1)
#Transposing the data chart to make plotting easier.GeeksForGeeks
#Row to name was utilized by library(janitor) to move barcode to name

t_eso_sub <- as.data.frame(t_eso_sub)
#Convert new matrix to df

t_eso_sub[] <- lapply(t_eso_sub, as.numeric)
#Convert char to num variables

#All Genes Selected for use in graphs and heat map:
#ENSG00000003989.18 SLC7A2 - *Main Selected Genes*
  #General Info - Maintains cellular uptake of cationic amino acids
#ENSG00000139514.13 SLC7A1 - Sister Gene Used for Scatter
#ENSG00000002079.14 MYH16
#ENSG00000005893.16 LAMP2
#ENSG00000001617.12 SEMA3F
#ENSG00000005001.10 PRSS22
#ENSG00000117461.15 PIK3R3
#ENSG00000117472.10 TSPAN1
#ENSG00000117533.15 VAMP4
#ENSG00000117543.21 DPH5
```

#Initial Data Summary

```
summary(t_eso_sub$ENSG00000003989.18)
```

```
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
##      6.0    971.5   5589.0  15161.7  17567.0 273027.0
```

```
#Summary analysis for selected gene SLC7A2
sd(t_eso_sub$ENSG00000003989.18)
```

```
## [1] 25880.15
```

```
#SD for selected gene SLC7A2

summary(t_eso_sub$ENSG00000139514.13)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     259    2129    3215    4003    4754   35938
```

```
#Summary analysis for 2nd gene SLC7A2
sd(t_eso_sub$ENSG00000139514.13)
```

```
## [1] 3241.912
```

```
#SD for 2nd gene SLC7A2

#Create Summary Table of SLC7A2 gene
SLC7A2_Stats <- data.frame(
  Mean = mean(t_eso_sub$ENSG00000003989.18),
  Median = median(t_eso_sub$ENSG00000003989.18),
  SD = sd(t_eso_sub$ENSG00000003989.18),
  Min = min(t_eso_sub$ENSG00000003989.18),
  Q1 = quantile(t_eso_sub$ENSG00000003989.18, 0.25),
  Q3 = quantile(t_eso_sub$ENSG00000003989.18, 0.75),
  Max = max(t_eso_sub$ENSG00000003989.18))

SLC7A2_Stats
```

```
##         Mean Median       SD Min    Q1    Q3    Max
## 25% 15161.68   5589 25880.15   6 971.5 17567 273027
```

```
#Use of knitr package to create summary table
knitr::kable(SLC7A2_Stats,
  digits = 2,
  caption = "Summary Table of SLC7A2 RNA-Seq Counts")
```

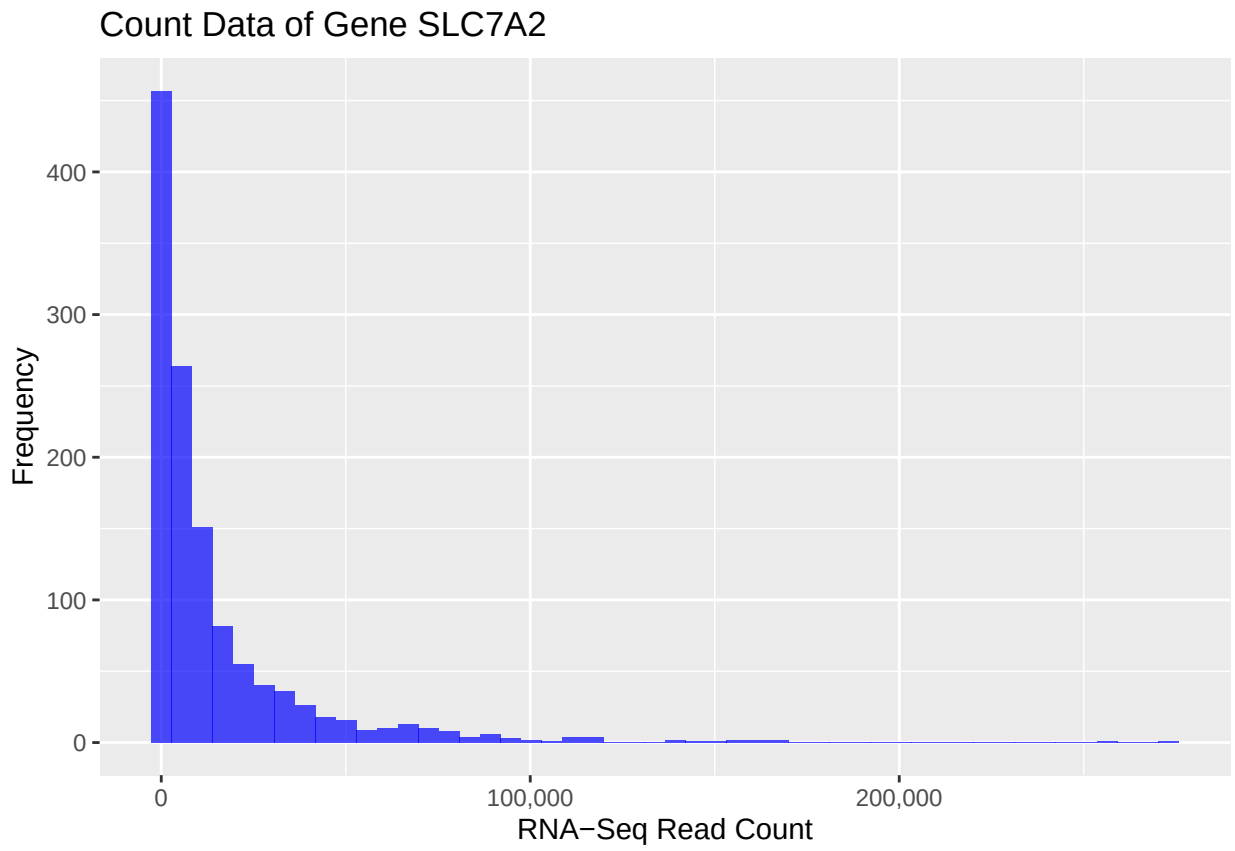
Table 1: Summary Table of SLC7A2 RNA-Seq Counts

	Mean	Median	SD	Min	Q1	Q3	Max
25%	15161.68	5589	25880.15	6	971.5	17567	273027

#Generate a histogram of the frequency of SLC7A2 read count

```
#Histogram
Histogram <- ggplot(t_eso_sub, aes(x = ENSG00000003989.18)) +
  geom_histogram(bins = 50, fill = "blue", alpha = 0.7) +
  scale_x_continuous(labels = scales::comma) +
  labs(title = "Count Data of Gene SLC7A2",
       x = "RNA-Seq Read Count",
       y = "Frequency")

Histogram
```

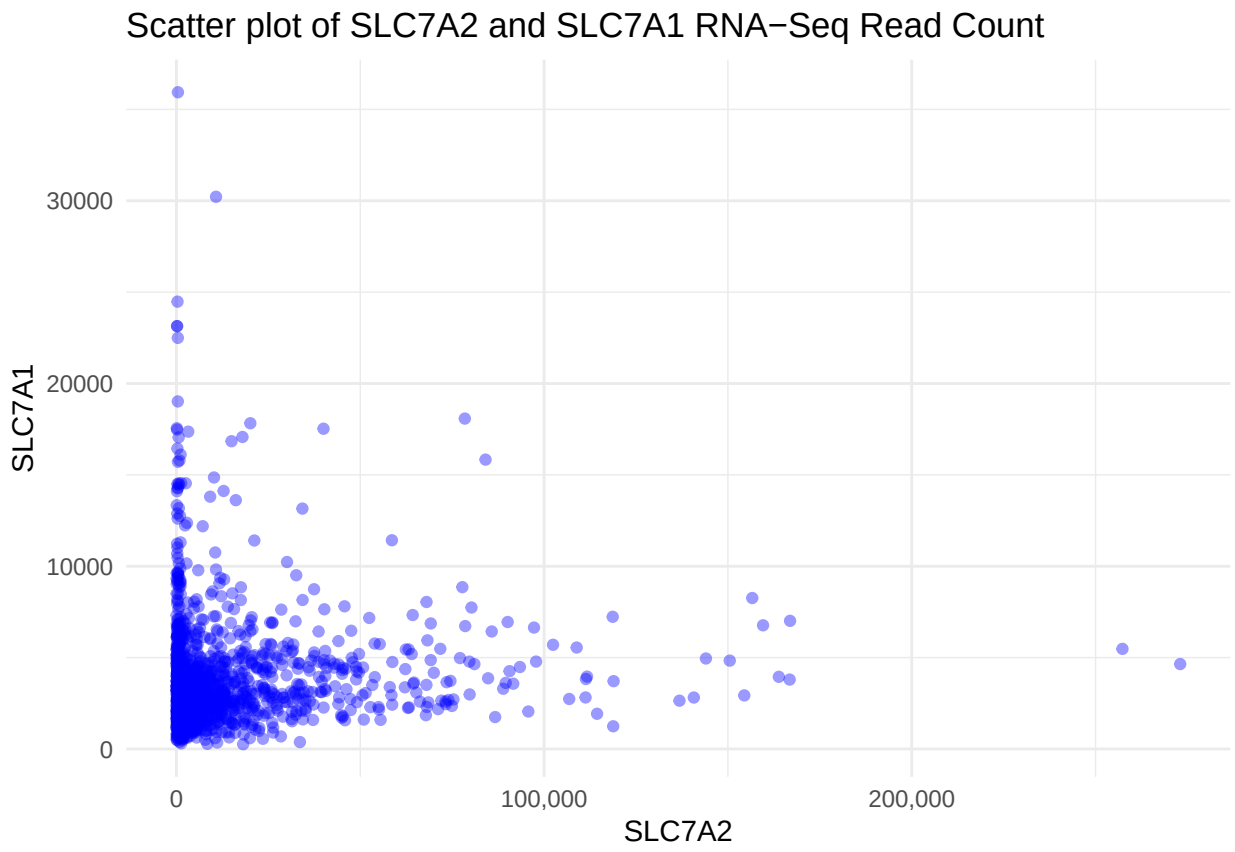



```
#Utilizing ggplot to generate a histogram based on the count
#of the main gene SLC7A2. The gene shows a heavy
#right-skew in the data.
```

#Generate a scatterplot between the SLC7A2 and SLC7A1 count

```r
#Scatterplot
Scatterplot <- ggplot(t_eso_sub,
                      aes(x = ENSG00000003989.18,
                          y = ENSG00000139514.13)) +
  geom_point(alpha = 0.4, color = "blue") +
  scale_x_continuous(labels = scales::comma) +
  labs(title = "Scatter plot of SLC7A2 and SLC7A1 RNA-Seq Read Count",
       x = "SLC7A2",
       y = "SLC7A1") +
  theme_minimal()

Scatterplot
```




```r
#The Scatterplot compares the two counts based on barcode and a relatively normal distribution, but ske
```

#Preparation and matching barcodes for Violin Plot

```r
t_eso_sub_m <- rownames_to_column(t_eso_sub, var = "barcode")
#Converts barcode to column to prepare for combining data frames based on barcode

meta_data$barcode <- gsub("-", ".", meta_data$barcode)
#Converts the "-" in the meta_data df to "." so the sample barcodes will match

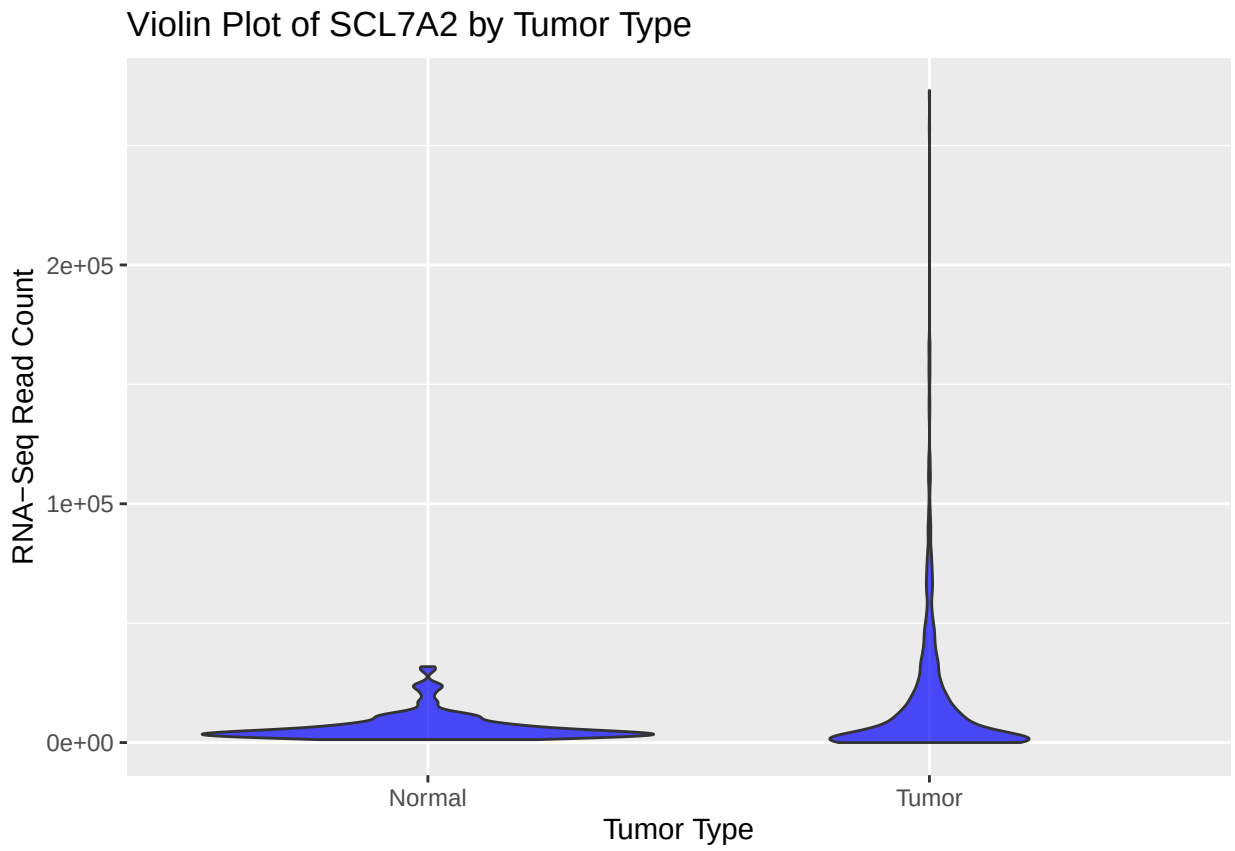
t_eso_sub_meta <- list(meta_data, t_eso_sub_m) %>%
```

```
  reduce(full_join, by = "barcode")
#Combines and creates a new data frame based on the barcodes
```

#Generate Violin Plot of SCLA

```
violin_plot <- ggplot(t_eso_sub_meta,
                      aes(x = tissue_type,
                          y = ENSG00000003989.18)) +
  geom_violin(fill = "blue", alpha = 0.7) +
  labs(title = "Violin Plot of SCL7A2 by Tumor Type ",
       x = "Tumor Type",
       y = "RNA-Seq Read Count")

violin_plot
```




```
#ggplot used to generate a violin plot. Visualizes RNA-seq count of the main gene
#based on whether the sample tissue is a tumor or normal.
```

#Preparing Heat Map

```
#Heatmap Df

library(ComplexHeatmap) #Packages to generate a heatplot
```

```
## Loading required package: grid

## ========================================
## ComplexHeatmap version 2.22.0
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##     genomic data. Bioinformatics 2016.
##
##
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(ComplexHeatmap))
## ========================================
```

```
library(circlize)
```

```
## Warning: package 'circlize' was built under R version 4.4.3

## ========================================
## circlize version 0.4.18
## CRAN page: https://cran.r-project.org/package=circlize
## Github page: https://github.com/jokergoo/circlize
## Documentation: https://jokergoo.github.io/circlize_book/book/
##
## If you use it in published research, please cite:
## Gu, Z. circlize implements and enhances circular visualization
##   in R. Bioinformatics 2014.
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(circlize))
## ========================================
```

```
genes_to_plot <- c("ENSG00000003989.18",
                   "ENSG00000002079.14",
                   "ENSG00000005893.16",
                   "ENSG00000139514.13",
                   "ENSG00000001617.12",
                   "ENSG00000005001.10",
                   "ENSG00000117461.15",
                   "ENSG00000117472.10",
                   "ENSG00000117533.15",
                   "ENSG00000117543.21")
#Makes list of genes to plot and select

counts_subset <- t_eso_sub_meta[, genes_to_plot ]
```

```r
#Generates new subset based on genes_to_plot list

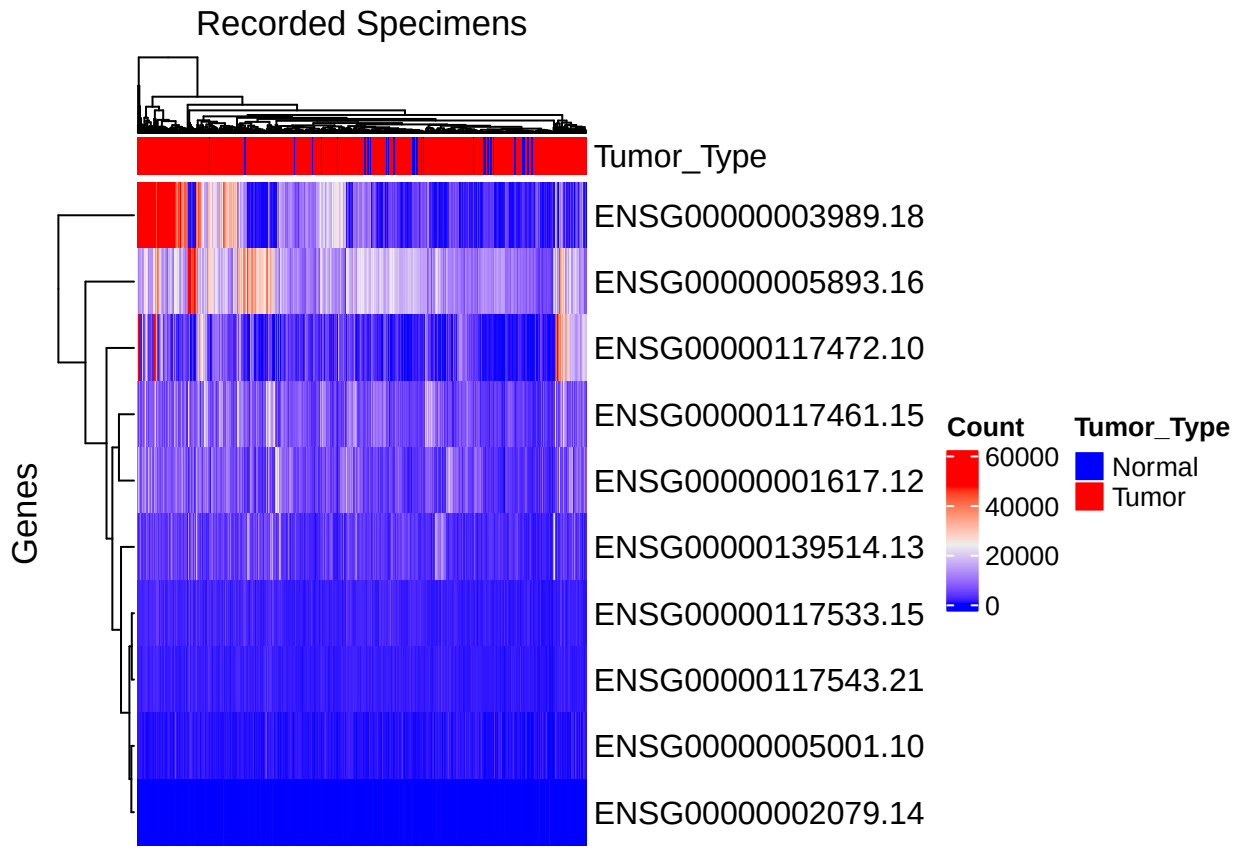
counts_subset <- t(as.matrix(counts_subset))
#Transposes the new subset as a matrix

column_ha <- HeatmapAnnotation(
  Tumor_Type = t_eso_sub_meta$tissue_type,
  col = list(
    Tumor_Type = c(Tumor = "red", Normal = "blue")
  ))
```

#Annotation of Heat Map Based on Tumor Type of Tumor or Normal

```r
Heatmap(
  counts_subset,
  name = "Counts",
  top_annotation = column_ha,
  show_row_names = TRUE,
  show_column_names = FALSE,
  cluster_columns = TRUE,
  cluster_rows = TRUE,
  column_title = "Recorded Specimens",
  row_title = "Genes",
  heatmap_legend_param = list(title = "Count"))
```

```
## The automatically generated colors map from the 1^st and 99^th of the
## values in the matrix. There are outliers in the matrix whose patterns
## might be hidden by this color mapping. You can manually set the color
## to ‘col‘ argument.
##
## Use ‘suppressMessages()‘ to turn off this message.
```




```
#Generates Heat Map of the selected genes in the subset based on the count
#and identified by whether the tissue sample is normal or tumor.
```

#Log-Transformed Heat Map

#The heat map is hard to interpret. Taking from my classmates notes, #I will do a log-transformation to help reduce the influence of #large counts.

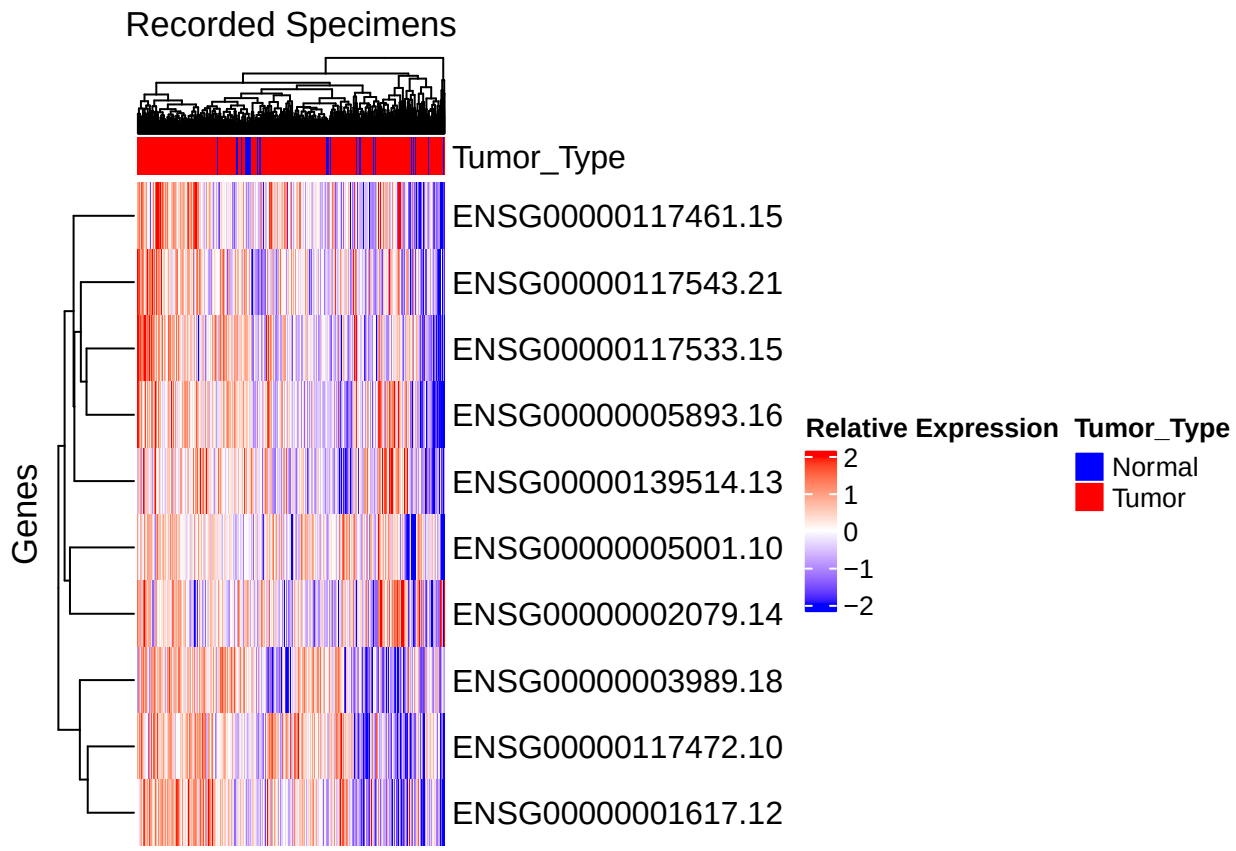
```
log_counts <- log2(counts_subset + 1)
#Log Transformation of data in subset

scaled_counts <- t(scale(t(log_counts)))
#Standardize the genes in samples

#Color Scale
col_fun <- circlize::colorRamp2(
  c(-2, 0, 2),
  c("blue", "white", "red"))

#Generate New Log Heatmap
Heatmap(
  scaled_counts,
  name = "Z-score",
  col = col_fun,
  top_annotation = column_ha,
  show_row_names = TRUE,
```

```
  show_column_names = FALSE,
  cluster_columns = TRUE,
  cluster_rows = TRUE,
  column_title = "Recorded Specimens",
  row_title = "Genes",
  heatmap_legend_param = list(
    title = "Relative Expression"))
```



#Preparing ROC Graph

```
library(pROC)
```

```
## Warning: package 'pROC' was built under R version 4.4.3
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```r
#Convert tumor status to a binary variable for ROC graph
t_eso_sub_meta$tumor_binary <- ifelse(
  t_eso_sub_meta$tissue_type == "Tumor", 1, 0)

#Converts tumor status to 1 and normal as 0

#As shown before, we recognize problems regarding RNA count difference
#between SCL7A2 and SCL7A1 and require log-transformation.

#Selects and transforms the variables for a new data frame
roc_df <- t_eso_sub_meta %>%
  filter(tissue_type %in% c("Normal", "Tumor")) %>% #Selects rows
  mutate(
    SLC7A2 = log2(ENSG00000003989.18 + 1), #Log Transformation
    SLC7A1 = log2(ENSG00000139514.13 + 1)) %>% #Forms new variables
  select(tumor_binary, SLC7A2, SLC7A1) %>% #Keeps selected variables
  drop_na() #Remove all missing data
```

#Model a generalized linear model using the tumor status #against the sister genes of SLC7A2 and SLC7A1

```r
model <- glm(
  tumor_binary ~ SLC7A2 + SLC7A1, #Use SLC7A2 and SLC7A1
  data = roc_df,
  family = binomial) #Selects binomial regression

#Generates results of model
summary(model)
```

```
##
## Call:
## glm(formula = tumor_binary ~ SLC7A2 + SLC7A1, family = binomial,
##     data = roc_df)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.78372    1.20968  -0.648  0.51707
## SLC7A2      -0.05629    0.03758  -1.498  0.13418
## SLC7A1       0.32667    0.10166   3.213  0.00131 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 755.03  on 1230  degrees of freedom
## Residual deviance: 742.85  on 1228  degrees of freedom
## AIC: 748.85
##
## Number of Fisher Scoring iterations: 5
```

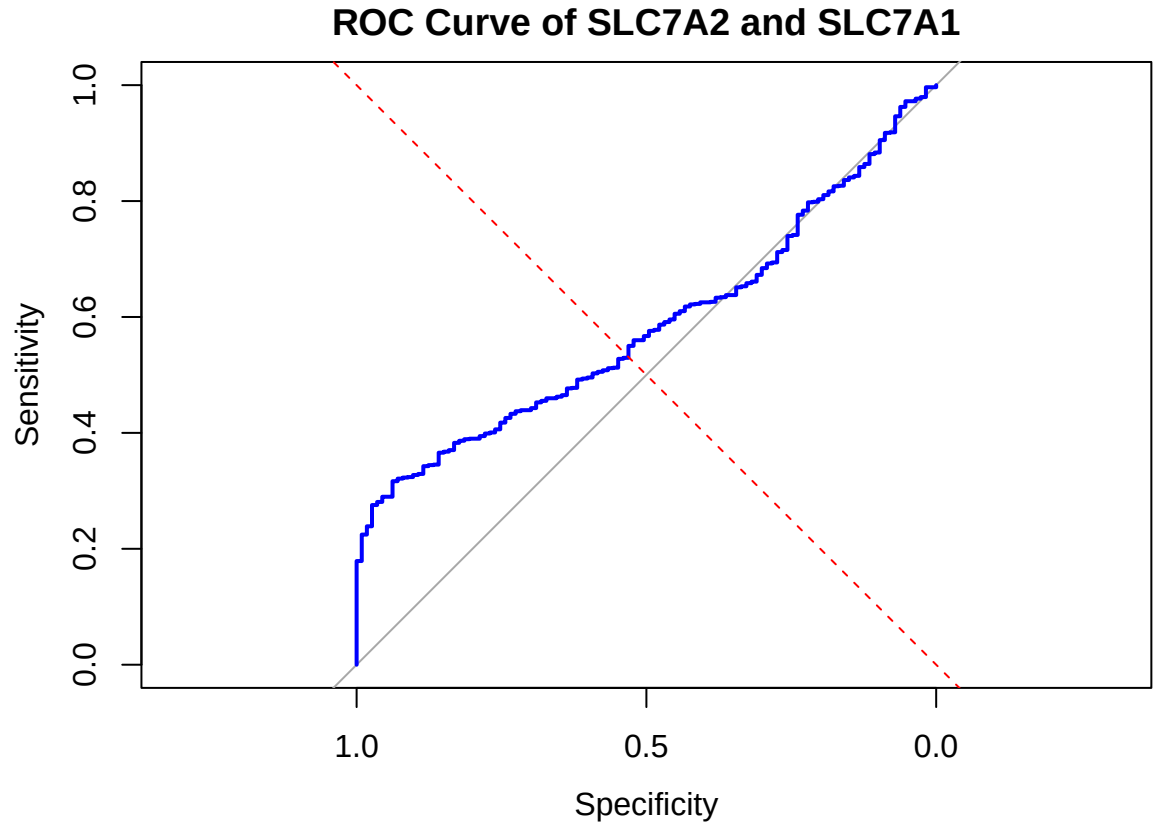
```r
#Creates prediction of tumor probability
roc_df$predicted <- predict(
  model,
```

```
  type = "response")

#Calculate ROC curve base using actual vs predicted
roc_curve <- roc(
  roc_df$tumor_binary,
  roc_df$predicted,
  levels = c(0,1),
  direction = "<")

plot(roc_curve, col = "blue", lwd = 2,
     main = "ROC Curve of SLC7A2 and SLC7A1")

abline(a = 0, b = 1, lty = 2, col = "red")
```




```
#Calculates Area Under Curve
auc(roc_curve)
```

```
## Area under the curve: 0.5836
```

```
#0.5836 is a very low ROC AUC, and those we can conclude that
#both SLC7A2 and SLC7A1 together are not effective in
#a predictive model for determining tumor status.
```